Systems Biology Project 2 Report

Spatiotemporal Signal Propagation

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Task A1: Mutation Comparison

This section compares the Wild Type (WT) control with several specific mutations (AKT1_E17K, PIK3CA_E545K, PIK3CA_H1047R, PTEN_del) to evaluate their impact on spatiotemporal ERK signaling. The Relative Risk (RR) metric was used to quantify the likelihood of a cell exhibiting an ERK activity spike given that a neighboring cell recently spiked.

A Mann-Whitney U test with Bonferroni correction was performed to compare the relative risk of each mutant against the WT. The results are summarized in Table 1.

Mutation	**Mean RR**	**Std Error**	**p-value vs WT**	**Significant**
AKT1_E17K	1.581	0.072	3.46×10^{-3}	True
PIK3CA_E545K	1.707	0.032	0.184	False
PIK3CA_H1047R	3.200	0.131	1.20×10^{-5}	True
PTEN_del	1.564	0.082	3.46×10^{-3}	True

Table 1: Mutation comparison for spatiotemporal signal propagation relative to WT.

Biological Interpretation: The PIK3CA_H1047R mutation significantly increases the Relative Risk compared to WT. This mutation acts at the plasma membrane, potentially promoting the secretion of paracrine factors or increasing sensitivity to neighbor-derived signals. This "hypersensitivity" facilitates efficient signal spread. In contrast, downstream mutations like AKT1_E17K and PTEN_del generate high constitutive baseline activity. This internal noise triggers negative feedback loops, desensitizing cells to external cues, which leads to asynchronous signaling and the observed drop in Relative Risk.

Task A2: Lagged Exposure

To understand the temporal dynamics of intercellular communication, we analyzed the Relative Risk across different time lags (τ) ranging from 0 to 30 minutes (0 to 6 frames) for the WT and two representative mutations: AKT1_E17K and PTEN_del. The objective was to determine if constitutive activation alters the optimal communication lag.

Lag τ (min)	Lag (frames)	WT RR	AKT1_E17K RR	PTEN_del RR
0	0	1.756	1.611	1.573
5	1	1.664	1.568	1.479
10	2	1.563	1.541	1.351
15	3	1.455	1.513	1.230
20	4	1.366	1.494	1.137
25	5	1.294	1.474	1.062
30	6	1.234	1.452	1.023

Table 2: Relative Risk ($RR^{(\tau)}$) across different exposure lags for WT, `AKT1_E17K`, and `PTEN_del`.

Biological Interpretation: For all cell lines, the Relative Risk is highest at $\tau = 0$ and steadily decreases as the lag increases. The lack of a shifted peak indicates that signal transmission is extremely rapid, with cells responding almost immediately to a neighbor's signaling event. The convergence of RR toward 1 at longer lags confirms that the neighbor effect is transient. Furthermore, WT cells exhibit the strongest response ($RR \approx 1.76$). The reduced coupling strength in `AKT1_E17K` and `PTEN_del` reflects their constitutively elevated internal activity, reducing their sensitivity to external cues.

Task A3: Parameter Robustness

The robustness of the spatial radius parameter (r) was evaluated for the WT cell line. We tested radii of 30, 60, 90, and 150 μm to determine the optimal distance for capturing genuine local signaling events without introducing excessive background noise.

Radius (μm)	Exposed Nodes	P(exp)	P(unexp)	Relative Risk (RR)
30	43,171	0.180	0.087	2.068
60	146,620	0.131	0.075	1.756
90	239,638	0.112	0.070	1.605
150	332,701	0.100	0.075	1.328

Table 3: Impact of spatial radius r on Relative Risk in WT cells.

Biological Interpretation: While a radius of 30 μm yields the highest Relative Risk ($RR \approx 2.07$), it captures very few neighbors per cell, leading to statistical volatility and the exclusion of genuine adjacent neighbors due to irregular cell shapes. Conversely, expanding the radius to 150 μm includes an enormous number of distant cells. This massive inclusion dilutes the measured effect as uncorrelated background noise overwhelms local signals (RR drops to 1.33). The chosen radius of 60 μm (spanning 1.5 to 2 epithelial cell diameters) is optimally aligned with the effective range of short-range paracrine diffusion, successfully capturing direct cell-to-cell communication.